VERSION HISTORY

Index:

P01 VERSION HISTORY
P02 BLOCK DIAGRAM
P03 POWER TREE
P04 GPIO ASSIGNMENT
P05 POWER
P06 RV1106G
P07 EMMC
P08 AUDIO
P09 BT&WIFI
P10 CAMERA
P11 DISPLAY
P12 CONNECTOR

OPTION:

Revision	Description	Date	Drawn	Checked	Approved
Ver 1.0	Release version	2025-09-02			

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Size A3	Document Number <Doc>	Rev <RevCode>
Date: Monday, August 03, 2026	Sheet 1 of 12	

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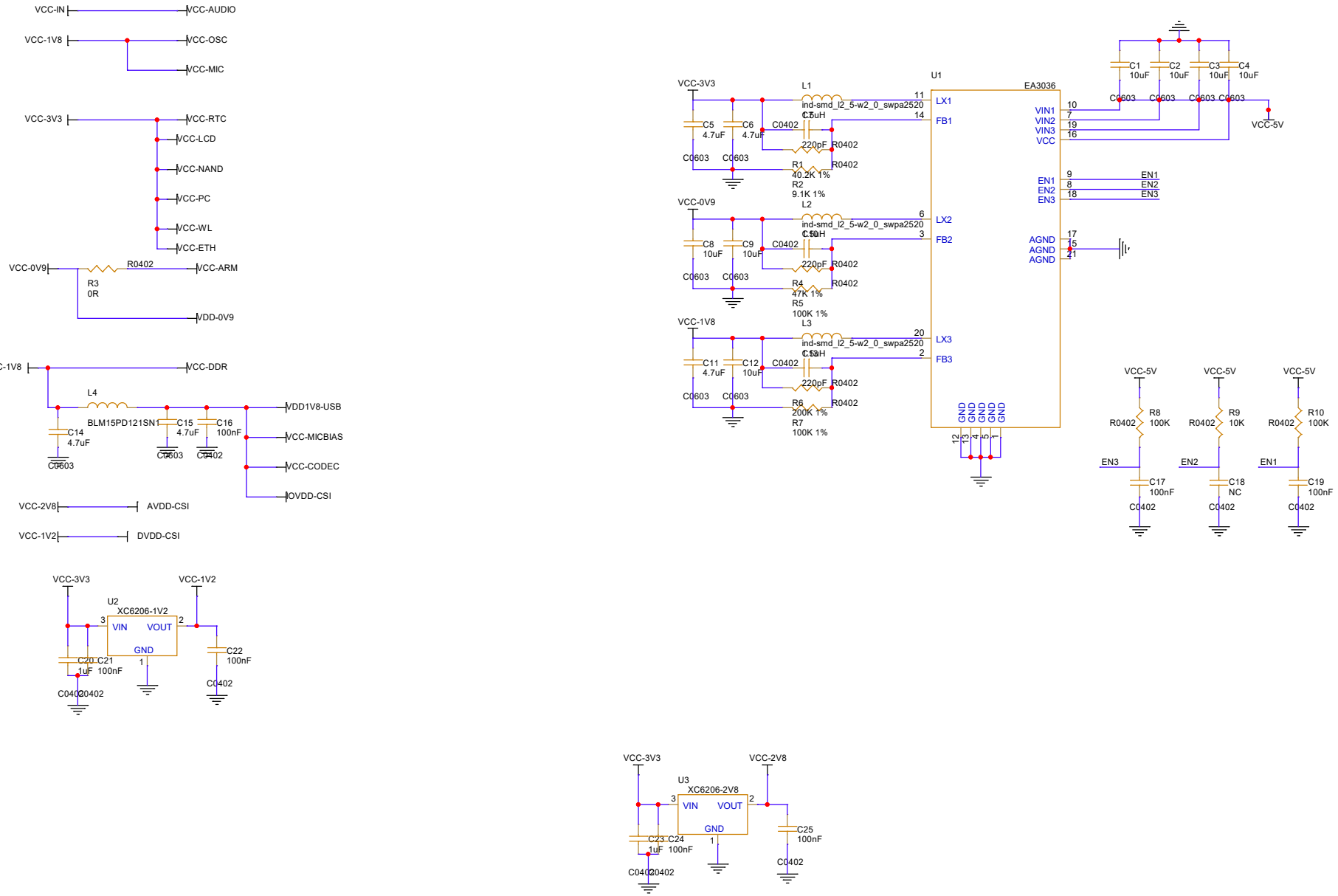
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Size A3	Document Number <Doc>	Rev <RevCode>
Date: Monday, August 03, 2026	Sheet 2 of 12	

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Size A3	Document Number <Doc>
Rev	<RevCode>
Date: Monday, August 03, 2026	Sheet 3 of 12

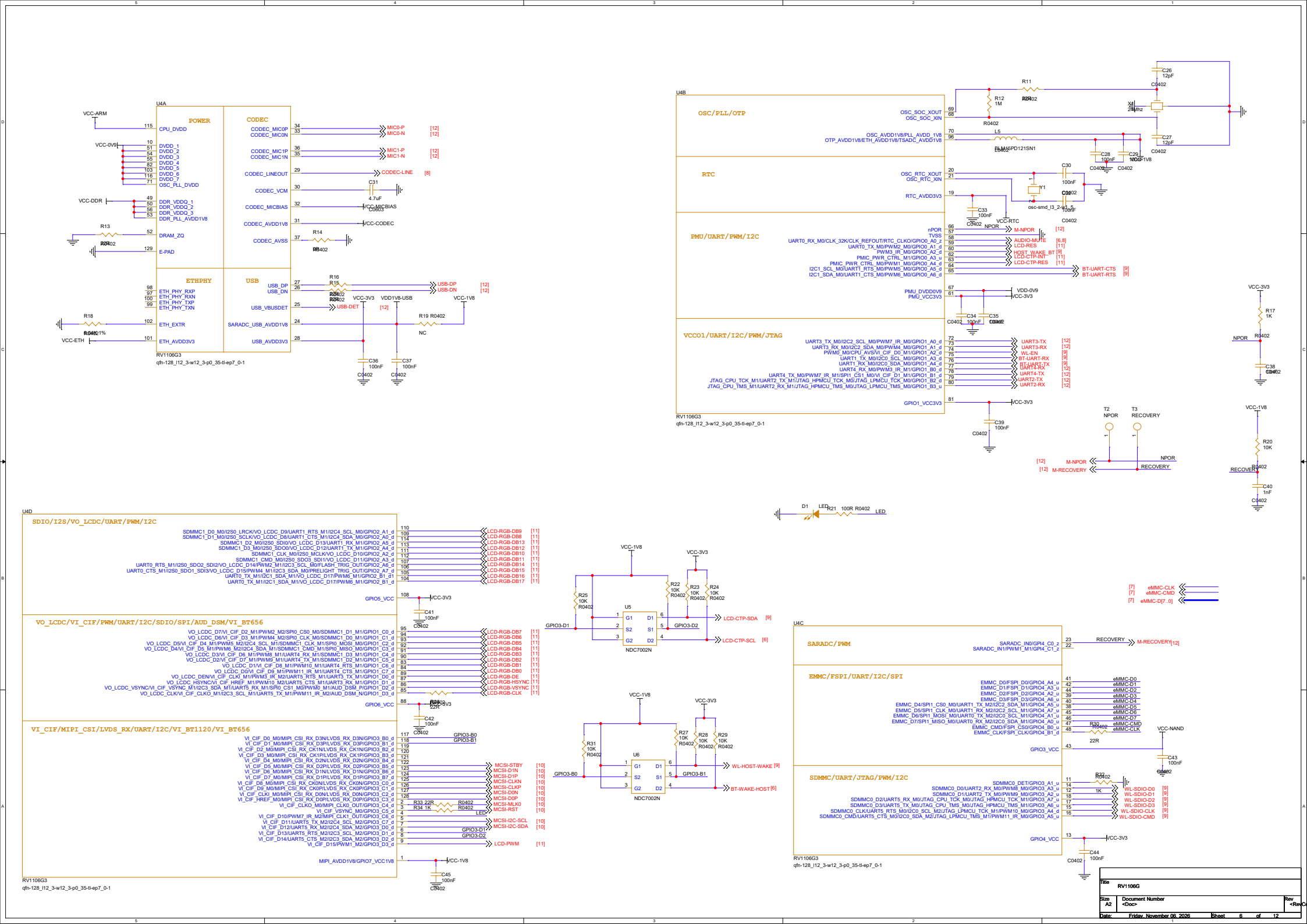
GPIO ASSIGNMENT

PIN	Define	CFG	Function
PB0	CPUX_TMS	4	DEBUG
PB1	CPUX_TCK	4	
PB2	CPUX_TDO	4	
PB3	CPUX_TDI	4	
PB4	I2S0_MCLK	3	I2S
PB5	I2S0_BCLK	3	
PB6	I2S0_LRCK	3	
PB7			
PB8	I2S0_DIN	3	
PB9	CPUX_TX	2	DEBUG
PB10	CPUX_RX	2	

Title	<Title>	
Size A3	Document Number <Doc>	Rev <RevCode>
Date: Monday, August 03, 2026	Sheet 4 of 12	



Title	<Title>		
Size A3	Document Number <Doc>		Rev <RevCode>
Date: Friday, November 06, 2026	Sheet 5 of 12		

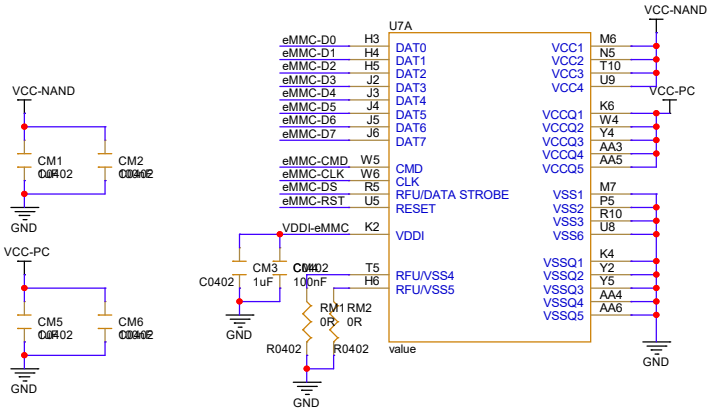

U4A
POWER
CPU_DVDD
DVDD_1
DVDD_2
DVDD_3
DVDD_4
DVDD_5
DVDD_6
DVDD_7
OSC_PLL_DVDD
DDR_VDDQ_1
DDR_VDDQ_2
DDR_VDDQ_3
DDR_PLL_AVDD1V8
DRAM_ZQ
E-PAD
CODEC
CODEC_MIC0P
CODEC_MIC0N
CODEC_MIC1P
CODEC_MIC1N
CODEC_LINEOUT
CODEC_VCM
CODEC_MICBIAS
CODEC_AVDD1V8
CODEC_AVSS
ETHPHY
ETH_PHY_RXP
ETH_PHY_RXN
ETH_PHY_TXP
ETH_PHY_TXN
ETH_EXTR
ETH_AVDD3V3
USB
USB_DP
USB_DN
USB_VBUSDET
SARADC_USB_AVDD1V8
USB_AVDD3V3
RV1106G3
qfn-128_l12_3-w12_3-p0_35-tl-ep7_0-1
U4B
OSC/PLL/OTP
OSC_SOC_XOUT
OSC_SOC_XIN
OSC_AVDD1V8/PLL_AVDD_1V8
OTP_AVDD1V8/ETH_AVDD1V8/TSADC_AVDD1V8
RTC
OSC_RTC_XOUT
OSC_RTC_XIN
RTC_AVDD3V3
PMU/UART/PWM/I2C
nPOR
TVSS
UART0_RX_M0/CLK_32K/CLK_REFOUT/RTC_CLKO/GPIO0_A0_z
UART0_TX_M0/PWM2_M0/GPIO0_A1_d
PWM3_IR_M0/GPIO0_A2_d
PMIC_PWR_CTRL_M1/GPIO0_A3_u
PMIC_PWR_CTRL_M0/PWM1_M0/GPIO0_A4_d
I2C1_SCL_M0/UART1_RTS_M0/PWM5_M0/GPIO0_A5_d
I2C1_SDA_M0/UART1_CTS_M0/PWM6_M0/GPIO0_A6_d
PMU_DVDD0V9
PMU_VCC3V3
VCCO1/UART/I2C/PWM/JTAG
UART3_TX_M0/I2C2_SCL_M0/PWM7_IR_M0/GPIO1_A0_d
UART3_RX_M0/I2C2_SDA_M0/PWM4_M0/GPIO1_A1_d
PWM0_M0/CPU_AVS/VI_CIF_D0_M1/GPIO1_A2_d
UART1_TX_M0/I2C0_SCL_M0/GPIO1_A3_d
UART1_RX_M0/I2C0_SDA_M0/GPIO1_A4_d
UART4_TX_M0/PWM3_IR_M1/GPIO1_B0_d
UART4_TX_M0/PWM7_IR_M1/SPI1_CS1_M0/VI_CIF_D1_M1/GPIO1_B1_d
JTAG_CPU_TCK_M1/UART2_TX_M1/JTAG_HPMCU_TCK_M0/JTAG_LPMCU_TCK_M0/GPIO1_B2_d
JTAG_CPU_TMS_M1/UART2_RX_M1/JTAG_HPMCU_TMS_M0/JTAG_LPMCU_TMS_M0/GPIO1_B3_u
GPIO1_VCC3V3
RV1106G3
qfn-128_l12_3-w12_3-p0_35-tl-ep7_0-1
U4D
SDIO/I2S/VO_LCDC/UART/PWM/I2C
SDMMC1_D0_M0/I2S0_LRCK/VO_LCDC_D9/UART1_RTS_M1/I2C4_SCL_M0/GPIO2_A1_d
SDMMC1_D1_M0/I2S0_SCLK/VO_LCDC_D8/UART1_CTS_M1/I2C4_SDA_M0/GPIO2_A0_d
SDMMC1_D2_M0/I2S0_SDI0/VO_LCDC_D13/UART1_RX_M1/GPIO2_A5_d
SDMMC1_D3_M0/I2S0_SDO0/VO_LCDC_D12/UART1_TX_M1/GPIO2_A4_d
SDMMC1_CLK_M0/I2S0_MCLK/VO_LCDC_D10/GPIO2_A2_d
SDMMC1_CMD_M0/I2S0_SDO3_SDI1/VO_LCDC_D11/GPIO2_A3_d
UART0_RTS_M1/I2S0_SDO2_SDI2/VO_LCDC_D14/PWM2_M1/I2C3_SCL_M0/FLASH_TRIG_OUT/GPIO2_A6_d
UART0_CTS_M1/I2S0_SDO1_SDI3/VO_LCDC_D15/PWM4_M1/I2C3_SDA_M0/PRELIGHT_TRIG_OUT/GPIO2_A7_d
UART0_TX_M1/I2C1_SDA_M1/VO_LCDC_D17/PWM6_M1/GPIO2_B1_d1
UART0_TX_M1/I2C1_SDA_M1/VO_LCDC_D17/PWM6_M1/GPIO2_B1_d
GPIO5_VCC
VO_LCDC/VI_CIF/PWM/UART/I2C/SDIO/SPI/AUD_DSM/VI_BT656
VO_LCDC_D7/VI_CIF_D2_M1/PWM2_M2/SPI0_CS0_M0/SDMMC1_D1_M1/GPIO1_C0_d
VO_LCDC_D6/VI_CIF_D3_M1/PWM4_M2/SPI0_CLK_M0/SDMMC1_D0_M1/GPIO1_C1_d
VO_LCDC_D5/VI_CIF_D4_M1/PWM5_M2/I2C4_SCL_M1/SDMMC1_CLK_M1/SPI0_MOSI_M0/GPIO1_C2_d
VO_LCDC_D4/VI_CIF_D5_M1/PWM6_M2/I2C4_SDA_M1/SDMMC1_CMD_M1/SPI0_MISO_M0/GPIO1_C3_d
VO_LCDC_D3/VI_CIF_D6_M1/PWM8_M1/UART4_RX_M1/SDMMC1_D3_M1/GPIO1_C4_d
VO_LCDC_D2/VI_CIF_D7_M1/PWM9_M1/UART4_TX_M1/SDMMC1_D2_M1/GPIO1_C5_d
VO_LCDC_D1/VI_CIF_D8_M1/PWM10_M1/UART4_RTS_M1/GPIO1_C6_d
VO_LCDC_D0/VI_CIF_D9_M1/PWM11_IR_M1/UART4_CTS_M1/GPIO1_C7_d
VO_LCDC_DEN/VI_CIF_CLKI_M1/PWM3_IR_M2/UART5_RTS_M1/UART3_TX_M1/GPIO1_D0_d
VO_LCDC_HSYNC/VI_CIF_HREF_M1/PWM10_M2/UART5_CTS_M1/UART3_RX_M1/GPIO1_D1_d
VO_LCDC_VSYNC/VI_CIF_VSYNC_M1/I2C3_SDA_M1/UART5_RX_M1/SPI0_CS1_M0/PWM0_M1/AUD_DSM_P/GPIO1_D2_d
VO_LCDC_CLK/VI_CIF_CLKO_M1/I2C3_SCL_M1/UART5_TX_M1/PWM11_IR_M2/AUD_DSM_N/GPIO1_D3_d
GPIO6_VCC
VI_CIF/MIPI_CSI/LVDS_RX/UART/I2C/VI_BT1120/VI_BT656
VI_CIF_D0_M0/MIPI_CSI_RX_D3N/LVDS_RX_D3N/GPIO3_B0_d
VI_CIF_D1_M0/MIPI_CSI_RX_D3P/LVDS_RX_D3P/GPIO3_B1_d
VI_CIF_D2_M0/MIPI_CSI_RX_CK1N/LVDS_RX_CK1N/GPIO3_B2_d
VI_CIF_D3_M0/MIPI_CSI_RX_CK1P/LVDS_RX_CK1P/GPIO3_B3_d
VI_CIF_D4_M0/MIPI_CSI_RX_D2N/LVDS_RX_D2N/GPIO3_B4_d
VI_CIF_D5_M0/MIPI_CSI_RX_D2P/LVDS_RX_D2P/GPIO3_B5_d
VI_CIF_D6_M0/MIPI_CSI_RX_D1N/LVDS_RX_D1N/GPIO3_B6_d
VI_CIF_D7_M0/MIPI_CSI_RX_D1P/LVDS_RX_D1P/GPIO3_B7_d
VI_CIF_D8_M0/MIPI_CSI_RX_CK0N/LVDS_RX_CK0N/GPIO3_C0_d
VI_CIF_D9_M0/MIPI_CSI_RX_CK0P/LVDS_RX_CK0P/GPIO3_C1_d
VI_CIF_CLKI_M0/MIPI_CSI_RX_D0N/LVDS_RX_D0N/GPIO3_C2_d
VI_CIF_HREF_M0/MIPI_CSI_RX_D0P/LVDS_RX_D0P/GPIO3_C3_d
VI_CIF_CLKO_M0/MIPI_CLK0_OUT/GPIO3_C4_d
VI_CIF_VSYNC_M0/GPIO3_C5_d
VI_CIF_D10/PWM7_IR_M2/MIPI_CLK1_OUT/GPIO3_C6_d
VI_CIF_D11/UART5_TX_M2/I2C4_SCL_M2/GPIO3_C7_d
VI_CIF_D12/UART5_RX_M2/I2C4_SDA_M2/GPIO3_D0_d
VI_CIF_D13/UART5_RTS_M2/I2C3_SCL_M2/GPIO3_D1_d
VI_CIF_D14/UART5_CTS_M2/I2C3_SDA_M2/GPIO3_D2_d
VI_CIF_D15/PWM1_M2/GPIO3_D3_d
MIPI_AVDD1V8/GPIO7_VCC1V8
RV1106G3
qfn-128_l12_3-w12_3-p0_35-tl-ep7_0-1
U4C
SARADC/PWM
SARADC_IN0/GPI4_C0_z
SARADC_IN1/PWM1_M1/GPI4_C1_z
EMMC/FSPI/UART/I2C/SPI
EMMC_D0/FSPI_D0/GPIO4_A4_u
EMMC_D1/FSPI_D1/GPIO4_A3_u
EMMC_D2/FSPI_D2/GPIO4_A2_u
EMMC_D3/FSPI_D3/GPIO4_A6_u
EMMC_D4/SPI1_CS0_M0/UART1_TX_M2/I2C2_SDA_M1/GPIO4_A5_u
EMMC_D5/SPI1_CLK_M0/UART1_RX_M2/I2C2_SCL_M1/GPIO4_A7_u
EMMC_D6/SPI1_MOSI_M0/UART0_TX_M2/I2C0_SCL_M1/GPIO4_A1_u
EMMC_D7/SPI1_MISO_M0/UART0_RX_M2/I2C0_SDA_M1/GPIO4_A0_u
EMMC_CMD/FSPI_CS0/GPIO4_B0_u
EMMC_CLK/FSPI_CLK/GPIO4_B1_d
GPIO3_VCC
SDMMC/UART/JTAG/PWM/I2C
SDMMC0_DET/GPIO3_A1_u
SDMMC0_D0/UART2_RX_M0/PWM8_M0/GPIO3_A3_u
SDMMC0_D1/UART2_TX_M0/PWM9_M0/GPIO3_A2_u
SDMMC0_D2/UART5_RX_M0/JTAG_CPU_TCK_M0/JTAG_HPMCU_TCK_M1/GPIO3_A7_u
SDMMC0_D3/UART5_TX_M0/JTAG_CPU_TMS_M0/JTAG_HPMCU_TMS_M1/GPIO3_A6_u
SDMMC0_CLK/UART5_RTS_M0/I2C0_SCL_M2/JTAG_LPMCU_TCK_M1/PWM10_M0/GPIO3_A4_d
SDMMC0_CMD/UART5_CTS_M0/I2C0_SDA_M2/JTAG_LPMCU_TMS_M1/PWM11_IR_M0/GPIO3_A5_u
GPIO4_VCC
RV1106G3
qfn-128_l12_3-w12_3-p0_35-tl-ep7_0-1
NDC7002N
NDC7002N
RECOVERY
M-RECOVERY
NPOR
M-NPOR
eMMC-CLK
eMMC-CMD
eMMC-D[7..0]
LED
WL-HOST-WAKE
BT-WAKE-HOST
LCD-PWM
LCD-CTP-SDA
LCD-CTP-SCL
MCSI-I2C-SCL
MCSI-I2C-SDA
WL-SDIO-D0
WL-SDIO-D1
WL-SDIO-D2
WL-SDIO-D3
WL-SDIO-CLK
WL-SDIO-CMD
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Size
A2
Document Number
<Doc>
Rev
<RevCo
Date:
Friday, November 06, 2026
Sheet
6
of
12

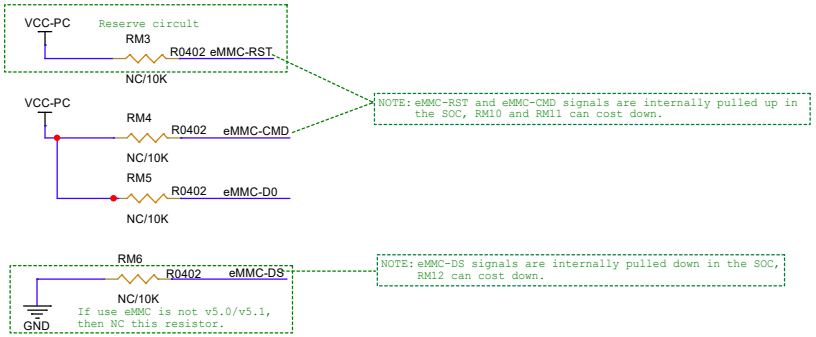
EMMC

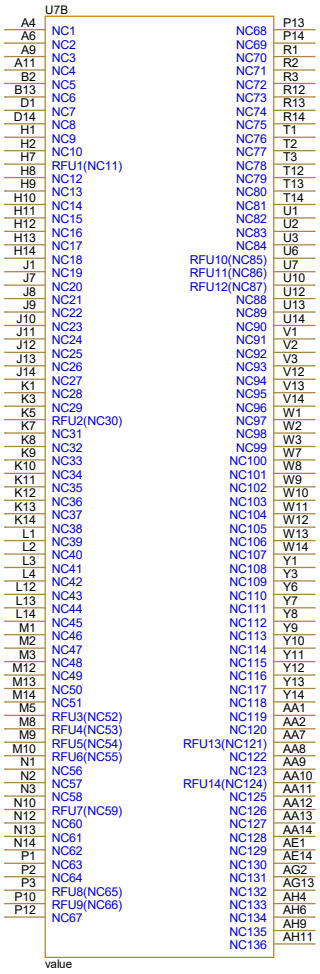
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[6] eMMC-CLK
[6] eMMC-CMD
[6] eMMC-D[7..0]

VCC-NAND 50mA
VCC-PC 100mA





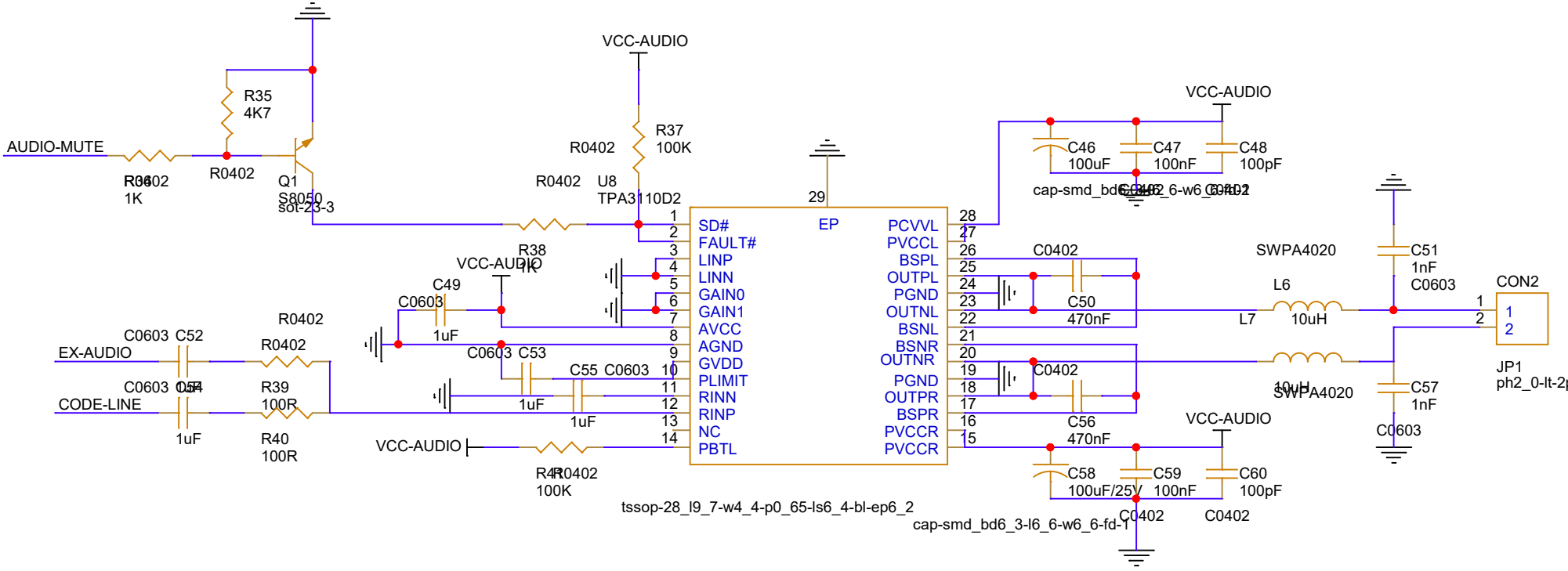


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Date: Monday, August 03, 2026	Sheet 7 of 12		

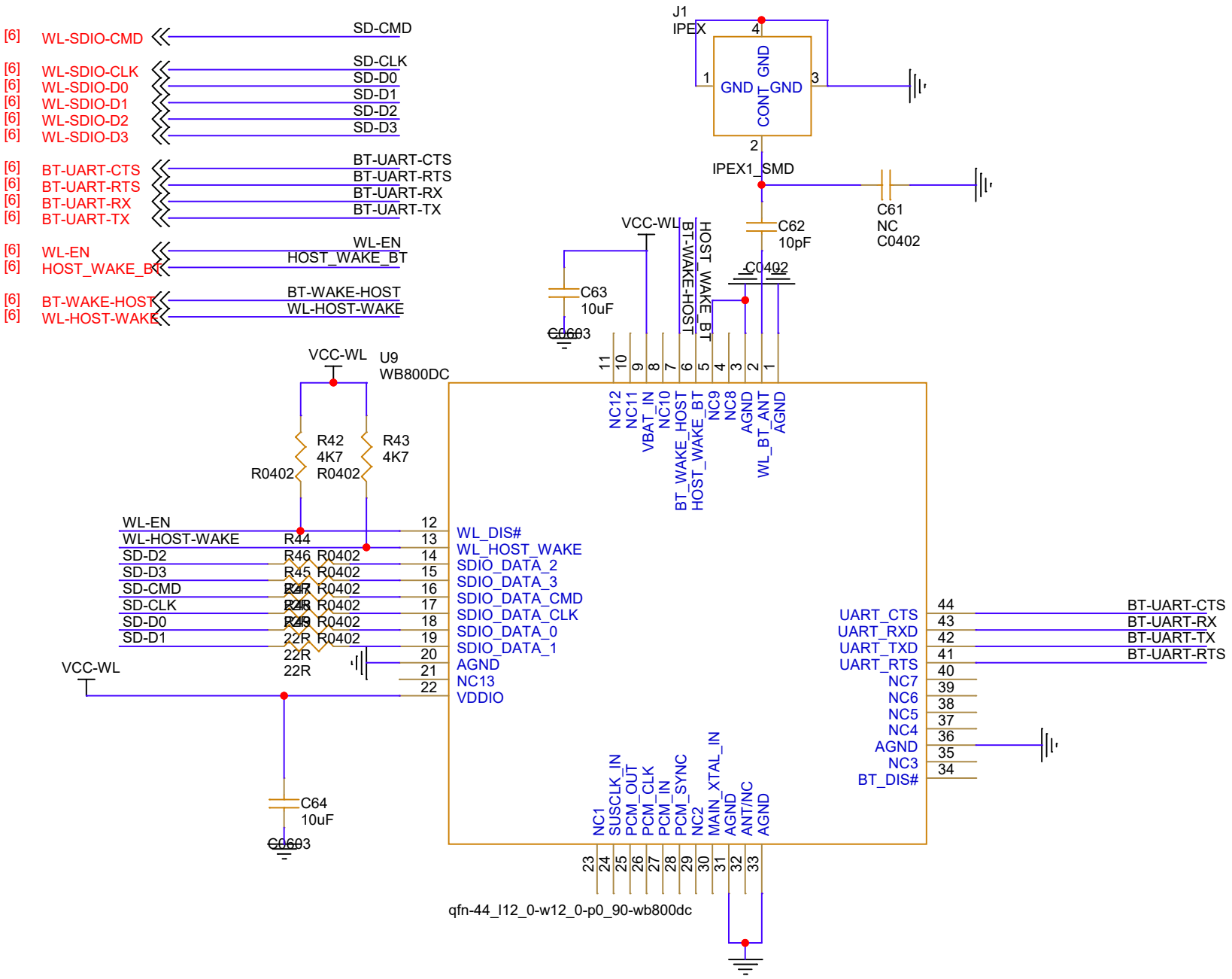
[12] EX-AUDIO <<< EX-AUDIO

[6] CODEC-LINE <<< CODE-LINE

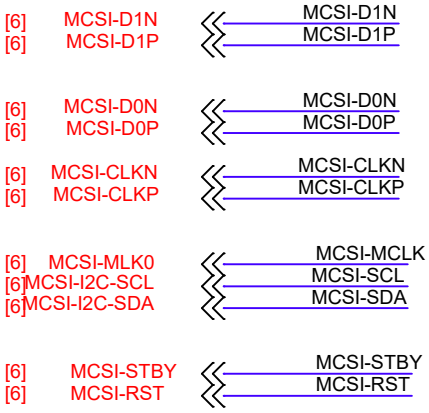
[6] AUDIO-MUTE <<< AUDIO-MUTE

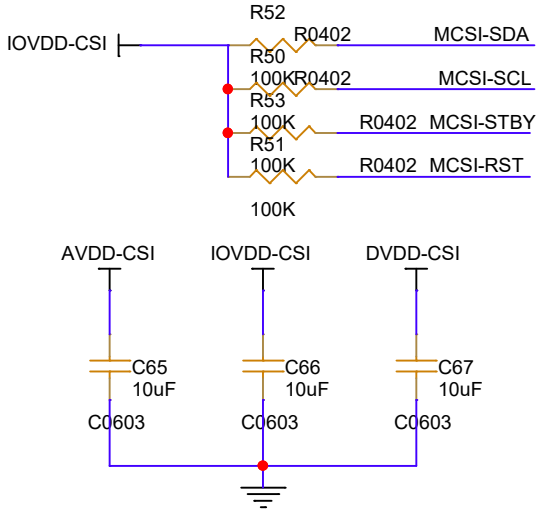


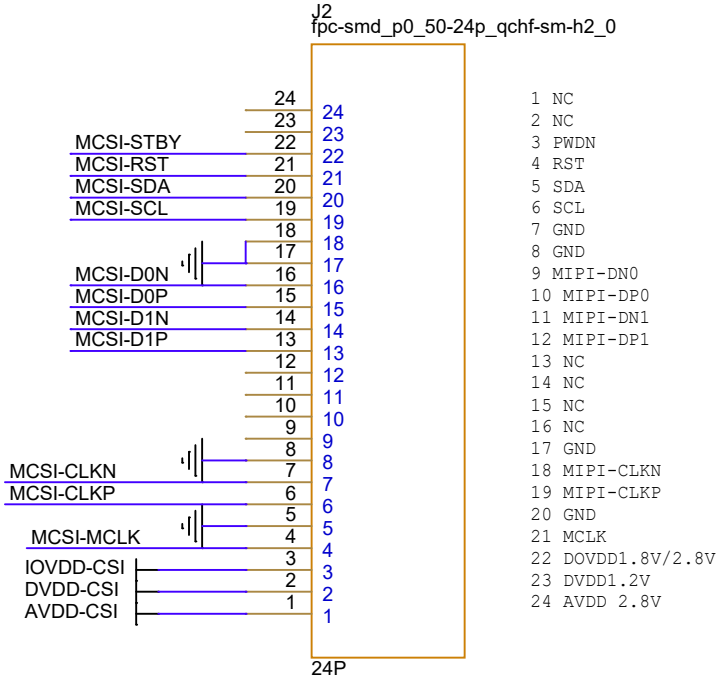
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Size A	Document Number <Doc>	Rev <RevCode>
Date: Monday, August 03, 2026	Sheet 8 of 12	



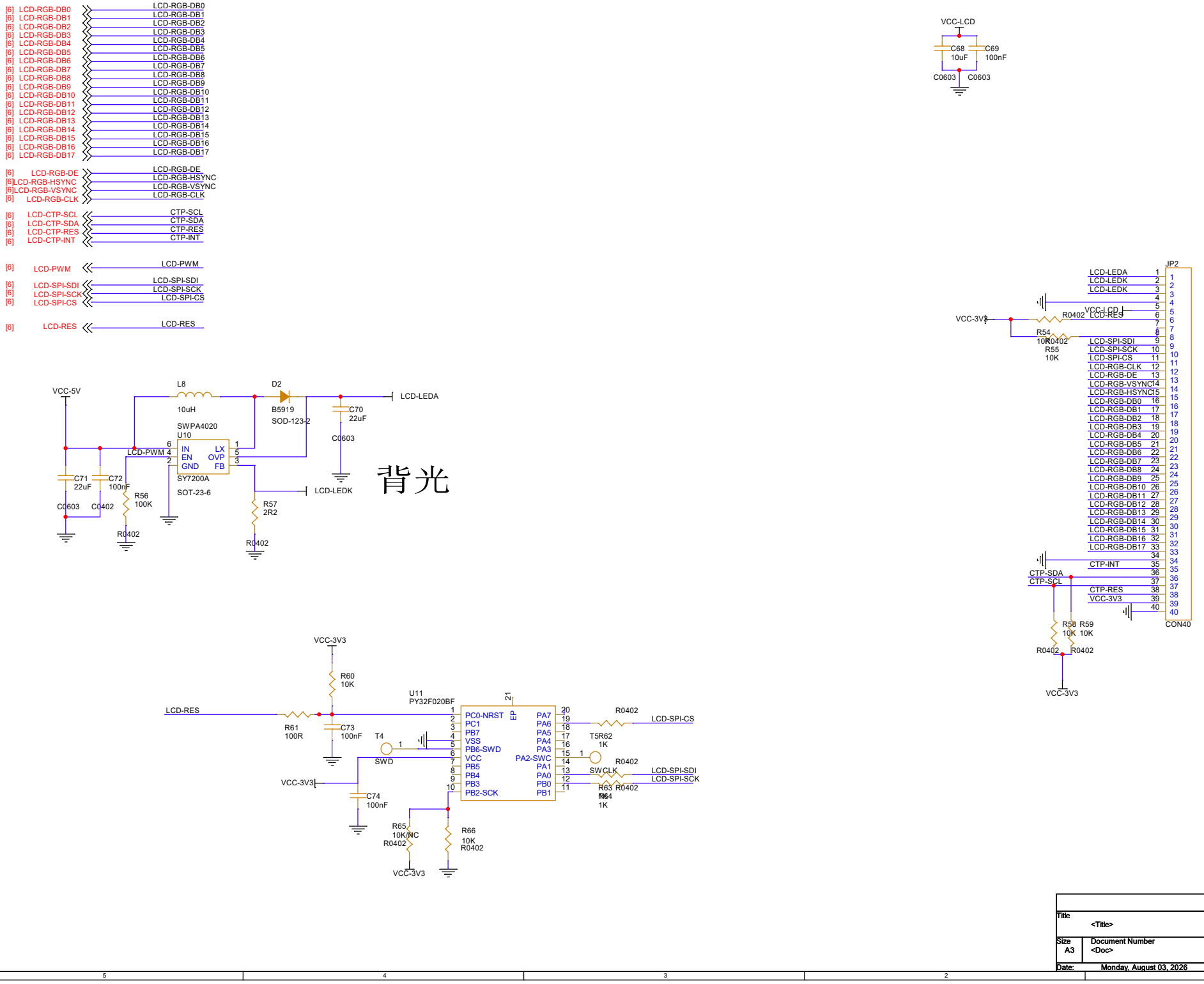
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Size A	Document Number <Doc>	Rev <RevCode>
Date: Monday, August 03, 2026	Sheet 9 of 12	







Title <Title>		
Size A	Document Number <Doc>	Rev <RevCode>
Date: Monday, August 03, 2026	Sheet 10 of 12	


[6] LCD-RGB-DB0 LCD-RGB-DB0
[6] LCD-RGB-DB1 LCD-RGB-DB1
[6] LCD-RGB-DB2 LCD-RGB-DB2
[6] LCD-RGB-DB3 LCD-RGB-DB3
[6] LCD-RGB-DB4 LCD-RGB-DB4
[6] LCD-RGB-DB5 LCD-RGB-DB5
[6] LCD-RGB-DB6 LCD-RGB-DB6
[6] LCD-RGB-DB7 LCD-RGB-DB7
[6] LCD-RGB-DB8 LCD-RGB-DB8
[6] LCD-RGB-DB9 LCD-RGB-DB9
[6] LCD-RGB-DB10 LCD-RGB-DB10
[6] LCD-RGB-DB11 LCD-RGB-DB11
[6] LCD-RGB-DB12 LCD-RGB-DB12
[6] LCD-RGB-DB13 LCD-RGB-DB13
[6] LCD-RGB-DB14 LCD-RGB-DB14
[6] LCD-RGB-DB15 LCD-RGB-DB15
[6] LCD-RGB-DB16 LCD-RGB-DB16
[6] LCD-RGB-DB17 LCD-RGB-DB17
[6] LCD-RGB-DE LCD-RGB-DE
[6] LCD-RGB-HSYNC LCD-RGB-HSYNC
[6] LCD-RGB-VSYNC LCD-RGB-VSYNC
[6] LCD-RGB-CLK LCD-RGB-CLK
[6] LCD-CTP-SCL CTP-SCL
[6] LCD-CTP-SDA CTP-SDA
[6] LCD-CTP-RES CTP-RES
[6] LCD-CTP-INT CTP-INT
[6] LCD-PWM LCD-PWM
[6] LCD-SPI-SDI LCD-SPI-SDI
[6] LCD-SPI-SCK LCD-SPI-SCK
[6] LCD-SPI-CS LCD-SPI-CS
[6] LCD-RES LCD-RES
VCC-LCD
C68 10uF
C69 100nF
C0603 C0603
VCC-5V
L8 10uH
D2 B5919 SOD-123-2
C70 22uF
C0603
LCD-LEDA
SWPA4020
U10
IN LX
EN OVP
GND FB
SY7200A
SOT-23-6
LCD-PWM
C71 22uF
C72 100nF
C0603 C0402
R56 100K
R0402
R57 2R2
R0402
LCD-LEDK
背光
JP2
LCD-LEDA 1
LCD-LEDK 2
LCD-LEDK 3
VCC-LCD
VCC-3V3
R0402
LCD-RES
R54 10K
R0402
R55 10K
LCD-SPI-SDI
LCD-SPI-SCK
LCD-SPI-CS
LCD-RGB-CLK
LCD-RGB-DE
LCD-RGB-VSYNC
LCD-RGB-HSYNC
LCD-RGB-DB0
LCD-RGB-DB1
LCD-RGB-DB2
LCD-RGB-DB3
LCD-RGB-DB4
LCD-RGB-DB5
LCD-RGB-DB6
LCD-RGB-DB7
LCD-RGB-DB8
LCD-RGB-DB9
LCD-RGB-DB10
LCD-RGB-DB11
LCD-RGB-DB12
LCD-RGB-DB13
LCD-RGB-DB14
LCD-RGB-DB15
LCD-RGB-DB16
LCD-RGB-DB17
CTP-INT
CTP-SDA
CTP-SCL
CTP-RES
VCC-3V3
CON40
R58 10K
R59 10K
R0402
R0402
VCC-3V3
VCC-3V3
R60 10K
LCD-RES
R61 100R
C73 100nF
U11
PY32F020BF
PC0-NRST
PC1
PB7
VSS
PB6-SWD
VCC
PB5
PB4
PB3
PB2-SCK
EP
PA7
PA6
PA5
PA4
PA3
PA2-SWC
PA1
PA0
PB0
PB1
T4
SWD
R0402
LCD-SPI-CS
T5R62 1K
SWCLK
R0402
LCD-SPI-SDI
LCD-SPI-SCK
R63 1K
R0402
VCC-3V3
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R0402
R66 10K
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Title
<Title>
Size A3
Document Number
<Doc>
Rev
<RevCode>
Date: Monday, August 03, 2026
Sheet 11 of 12

Master Board

[6] UART4-RX
[6] UART4-TX
MCU-TX1
MCU-RX1

[6] UART3-RX
[6] UART3-TX
MODULE-TX
MODULE-RX

[8] EX-AUDIO
EX-AUDIO-P

[6] MIC0-P
[6] MIC0-N
MIC0-P
MIC0-N

[6] MIC1-P
[6] MIC1-N
MIC1-P
MIC1-N

[6] UART2-RX
[6] UART2-TX
UART2-RX
UART2-TX

[6] MCU-REST
[6] M-RECOVERY
[6] M-NPOR
MCU-REST
M-RECOVERY
M-NPOR

[6] USB-DP
[6] USB-DN
M-USB-DP
M-USB-DN

[12] USB-DET
USB-DET

U12
MIC1-P 1
MIC1-N 2
MIC0-P 4
MIC0-N 5
VCC-MIC 8
1 2 3 4 5 6 7 8
CON8

JP3
VCC-5V 1
M-USB-DP 2
M-USB-DN 3
4
1 2 3 4
CON4

JP4
VCC-3V3 1
UART2-RX 2
UART2-TX 3
4
1 2 3 4
CON4

CON24
1 VCC-IN
2
3
4
5 VCC-5V
6
7 M-USB-DP
8 M-USB-DN
9
10 UART2-RX
11 UART2-TX
12 M-RECOVERY
13 M-NPOR
14 USB-DET
15 MCU-TX1
16 MCU-RX1
17 MODULE-TX
18 MODULE-RX
19 EX-AUDIO-P
20
21
22
23
24
J3

M1
1.0

M2
1.0

M3
1.0

M4
1.0

Title	<Title>	
Size A4	Document Number <Doc>	Rev <RevCo
Date: Monday, August 03, 2026	Sheet 12 of 12	